

AI-BASED ENTERPRISE KNOWLEDGE ASSISTANT

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Abstract - Accessing accurate administrative and faculty details in educational institutions is frustrating and time-consuming exercise. As department locations, room numbers, faculty office addresses and contact details are distributed across physical notice boards, website pages and PDF circulars, students- particularly new ones- waste extensive time searching on campus for these details. We present an Enterprise Knowledge Assistant to consolidate this information to a single interface to allow users to search these details in natural everyday language and receive accurate and targeted answers. The system is built with institutional data structured as high-dimensional vector embeddings and stored in a vector database to allow efficient retrieval aided by Retrieval-Augmented Generation (RAG) and semantic searches to ground answers into verified institutional data. As an instant knowledge repository, the assistant reduces search time of students and reduces repetitive queries to administrative staff, improving daily campus operations. This work showcases how domain-specific Retrieval-Augmented Generation systems can aid institutional knowledge access and enhance the student experience.

Keywords: Enterprise Knowledge Assistant, Retrieval-Augmented Generation (RAG), Semantic Search, Natural Language Processing (NLP), Vector Embeddings, Large Language Models (LLMs).

I.INTRODUCTION

Finding administrative and faculty details in a large educational institution can be frustrating and time-consuming exercise. Students- particularly new ones- are forced to spend long hours searching campus to find information about office addresses, room numbers or contact details of faculty. This information is stored across physical notice boards, embedded in the website or distributed as PDF circulars- making the search task even more challenging. The burden of understanding the administrative structure of the institution falls upon the shoulders of new students and is exacerbated by a lack of centralized search resources. While portals are digital, keyword searching often fails to return direct results and results in long lists of documents, links or PDFs for the user to download and sift through to find a particular answer.

The problem arises from the need for an enterprise-level knowledge repository that can understand natural language queries to allow for immediate, targeted answers. Recent advances in Artificial Intelligence (AI) and Natural Language Processing (NLP) have enabled conversational agents as information retrieval tools but standalone Large Language Models (LLMs) have a disadvantage of being "hallucinating": generating answers based on the training data even if it does not exist in the database.

Retrieval-Augmented Generation (RAG) overcomes this by using a pipeline that connects a document database containing verified institutional details with a LLM to generate answers that are grounded in the database.

This paper presents an AI-Based Enterprise Knowledge Assistant for institutional information retrieval. Institutional administrative details and faculty registers are consolidated into a centralized knowledge base that is structured as high-dimensional vector embeddings for efficient semantic retrieval. When a user asks a question in natural language, an embedding of the question is generated and used to perform semantic search and identify the most relevant context from the database to construct a RAG-based pipeline to answer the query.

By eliminating the need to sift through PDFs or search campus, the proposed assistant reduces search time for both students and administrative staff. The rest of this paper is organized as follows: section II is a literature review on RAG-based chatbots, section III describes the methodology and system design, section IV details the implementation and performance analysis and finally, section V concludes with future improvements.

II. LITERATURE REVIEW

Retrieval-Augmented Generation (RAG) has become a popular approach for developing knowledge-intensive conversational systems by combining information retrieval with generative

language models. Lewis et al. [1] introduced the RAG architecture, where relevant information is retrieved from an external knowledge source before generating the final response. This approach helps language models use additional information while answering questions and reduces their dependence on only the knowledge stored within the model.

Swacha and Gracel [2] presented a survey on the use of RAG-based chatbots in education. Their study discussed how RAG can be used to provide students with information from educational resources and institutional documents. The use of retrieved information was found to be useful for reducing unsupported or hallucinated responses. However, the study mainly covers the broader use of RAG in education and does not specifically focus on retrieving detailed campus administrative information.

Lee [3] proposed a RAG-based intelligent college information question-and-answer system. The system was designed to provide users with college-related information through natural language questions. The study demonstrated that RAG can be used to connect college information sources with a conversational interface and provide more relevant answers. However, there is still scope for improving the retrieval of highly specific information such as faculty contacts, room numbers, office locations and departmental details.

A study published in IJRASET [4] developed a RAG-based AI chatbot for student and institutional assistance using technologies such as semantic embeddings and local language models. The system aimed to reduce the effort

required by students to search for common institutional information manually. The study showed that RAG can be applied to automate regular student queries using institutional documents. However, a more focused approach is required for retrieving specific administrative and location-based information from different campus resources.

Neupane et al. [5] developed an informed chatbot for university resources to help students obtain information through natural language queries. The system focused on retrieving relevant university information and providing answers without requiring students to manually search through different resources. The study showed that retrieval-based approaches can improve access to university information. However, the system mainly focused on general university resources, leaving scope for more detailed retrieval of specific administrative information such as room numbers, faculty offices and department locations.

Olawore et al. [6] proposed a university chatbot using deep learning and a Retrieval-Augmented Generation approach. The study explored how RAG can be used to provide students with relevant answers from university information sources while reducing the possibility of unsupported responses. The work demonstrated the usefulness of combining language models with retrieved institutional information. However, the focus was mainly on chatbot-based student support, with less emphasis on structured relationships between departments, faculty members, rooms and administrative offices.

Nguyen et al. [7] investigated the integration of information retrieval techniques with Large Language Models for developing a document retrieval chatbot in an educational environment. Their work showed that retrieving relevant documents before generating an answer can improve the reliability of responses. The approach is useful for handling information stored across multiple educational documents. However, document-based retrieval alone may not be sufficient for queries that require exact relationships between different entities, such as identifying the department of a faculty member or locating a particular office on campus.

Alwi Yafie et al. [8] developed an academic chatbot using Retrieval-Augmented Generation for providing campus information services. The study demonstrated how RAG can be used to retrieve academic and institutional information and generate responses in a conversational manner. This approach can reduce the need for students to search through multiple sources manually. However, the system mainly addresses general campus information, while more focused retrieval of administrative details and location-specific information can still be improved.

Harsha Vardhan et al. [9] proposed AskUni, a RAG-based framework designed for intelligent university information systems. The system uses retrieval and generation techniques to provide users with relevant university information through natural language interaction. The study highlights the potential of RAG for making university information easier to access and reducing dependence on manual searches. However, the framework provides a general university information solution and does not

specifically concentrate on connecting faculty, departments, rooms and administrative entities for more precise institutional queries.

Wu and Tseng [10] developed a local generative AI teaching assistant using Retrieval-Augmented Generation technology to support the campus learning environment. The study focused on using locally processed institutional information with a generative AI system. This approach can help provide responses based on specific educational resources rather than relying only on general model knowledge. However, its primary focus is on teaching and learning support rather than administrative and campus-navigation queries such as faculty contacts, office locations and department details.

Peyton et al. [11] reviewed the use of university chatbots for student support and discussed how these systems have developed beyond simple frequently asked questions. The review highlights the use of chatbots for providing students with faster access to university services and information. It also shows that conversational systems can reduce the effort required to find common institutional information. However, traditional chatbot systems may face difficulties when users ask detailed or less predictable questions, creating a need for systems that can retrieve information from updated institutional documents

The systematic survey presented in [12] examined the use of Retrieval-Augmented Generation in educational applications. The study discussed different applications of RAG and highlighted its ability to combine information retrieval with generative models. RAG allows

responses to be generated using information retrieved from external knowledge sources instead of depending only on the knowledge stored in the language model. The study also highlights the importance of retrieval quality and reliable knowledge sources for generating useful responses. This supports the need for a properly organized institutional knowledge base in campus information systems.

Hayrapetyan [13] investigated the evaluation of Retrieval-Augmented Generation in a university policy chatbot. The study focused on evaluating RAG responses without depending completely on manually labelled datasets. This is useful in university environments where policy documents contain large amounts of information and may be updated over time. The work highlights the importance of checking whether generated responses are supported by the retrieved university information. However, the study mainly focuses on university policy information, whereas the proposed system also handles practical institutional details such as faculty contacts, department locations and room information.

Kavithra et al. [14] proposed a customized RAG support system for educational institutes. The study demonstrates how RAG can be adapted to the specific information requirements of an educational organization instead of using a general-purpose chatbot. By using institutional information as the knowledge source, the system can provide more relevant responses to users. However, there is still scope for improving the organization and retrieval of highly specific campus information by combining semantic search with structured databases and

relationships between different institutional entities.

Khan et al. [15] developed an educational virtual assistant using Large Language Models and a Retrieval-Augmented Generation framework to provide information support to university students. The system retrieves relevant information from university sources before generating responses to queries related to areas such as admission processes, course selection and campus facilities. The study demonstrated that combining university knowledge sources with RAG can provide more relevant and updated responses to student queries. However, the system mainly focuses on general university assistance, while more specific institutional queries involving faculty contacts, department rooms, office locations and relationships between campus entities can be further addressed.

III.METHODOLOGY

The proposed system utilizes a Retrieval-Augmented Generation (RAG) architecture with a dual database design to answer natural language questions regarding campus faculty and administrative departments.

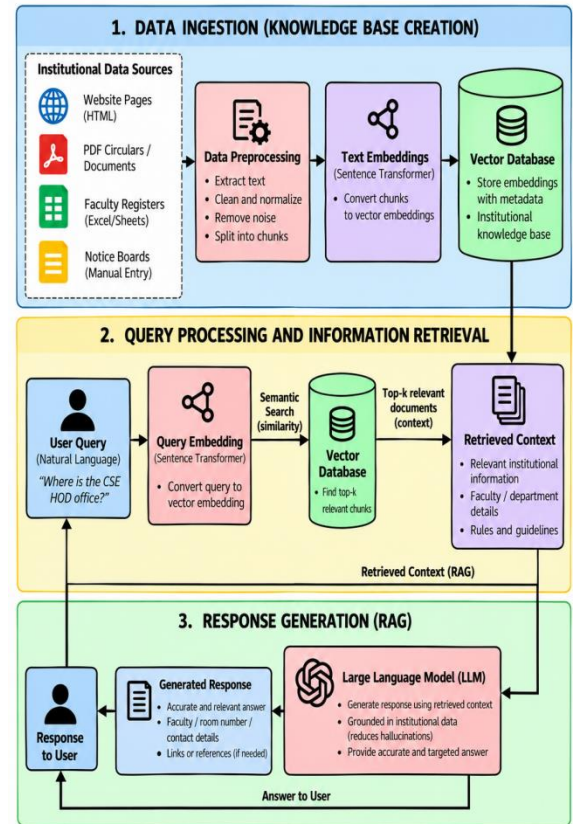


Figure 3.1. Architecture Diagram of the system

A. Data Ingestion & Indexing

The system is built using administrative details, such as departmental addresses and room numbers, and faculty contact details structured as entity tuples. The overall architecture is shown in Fig. 3.1. The validated dataset is stored in a relational Master Database to allow precise attribute lookup, as well as in a Document Store. Key textual attributes are also transformed into high dimensional vector embeddings and stored in a Vector Database to allow semantic search. To preserve structural context, such as faculty-department affiliations and physical campus layouts, entity relationships are mapped to an interconnected Knowledge Graph to aid in relational queries.

B. Semantic Retrieval & Response Pipeline

When a user asks a question through the web or mobile application, the API layer encodes the query into a vector. The system performs a hybrid search of semantic similarities from the vector database and graph-traversal through the Knowledge Graph to extract exact relational details. The extracted context is sent through the RAG pipeline and synthesizes a direct answer with the relevant room or contact details.

IV. DATA INGESTION MODULE

The Data Ingestion Module functions as the backbone of the Enterprise Knowledge Assistant, gathering and organizing scattered campus information. The main function of this module is to consolidate raw institutional data such as department directories, room locations, phone extensions, faculty rosters and office hours into a single knowledge base.

To ensure accuracy in the system, the raw records are cleaned and standardized before being converted into a database. Outdated and duplicated records are filtered to maintain the integrity of the system as the vector retrieval and generative AI components rely on the accuracy of the data set.

By collating disconnected administrative files and notice board updates, this module eliminates the hassle of manual campus searches. The cleaned data is formatted for vector embedding and semantic indexing to prepare it for instant retrieval. In this way, the module connects raw institutional records to the vector database and RAG system, allowing students to receive accurate answers to their everyday queries.

V. RAG RESPONSE GENERATION MODEL

The RAG Response Generation Module transforms retrieved institutional data into clear and direct answers. Following vector database isolation through semantic search, this module packages the details and the original natural language student query into context for the language model.

By restricting the context to the extracted information, the module synthesizes direct and accurate answers rather than generic summaries. For instance, if a student asks "Where is the Examination Cell?", the module extracts the specific room record and generates an answer of "Examination Cell is located in Room 204".

This contextual grounding eliminates the hassle of downloading and reading through multi-page policy PDFs, just to find a single address. Acting as the bridge between semantic retrieval and the user interface, the response generation module delivers instant responses to students.

VI.RESULT AND ANALYSIS

Query Category	Input (Student Natural-Language Query)	Output (Generated RAG Response)
Faculty Location	Where is Dr. Rajesh K. Sharma's cabin located and what is his email address	Dr. Rajesh K. Sharma's office is located in Admin Block, 3 rd Floor. His office hours are Monday – Friday from 10:00 AM – 12:00 PM His Email Address is Rajesh.sharma@rajalakhmi.edu.in.
Administrative Support	Which office handles official transcript requests and where are they?	Transcript requests are processed by the Academic Affairs Office, located in the Main Administration Building, Room 102.
Contact Retrieval	How can I contact the Head of the Electrical Engineering Department?	You can reach the HOD of Electrical Engineering via email at ee.hod@institution.edu or call extension 4102 (Room 215, EE Block).
Service Inquiries	Where do I submit	Fee waiver forms must be submitted to the Student Accounts

	my fee waiver form?	Desk in Room 108 of the Administrative Wing.
Exam Cell Services	Where do I apply for a duplicate hall ticket?	Duplicate hall tickets are issued at the Controller of Examinations Counter in the Exam Cell, Ground Floor, Central Block.
Library & Resources	Who is in charge of the digital library access and what is their email?	Digital library access is managed by Mr. Ramesh Kumar (Assistant Librarian). You can contact him at ramesh.lib@institution.edu.
Laboratory Location	Where is the Cyber Security and Digital Forensics Laboratory located and what tools does it have?	The laboratory is housed within the Department of Computer Science and Engineering. The laboratory is equipped with isolated network environments used for vulnerability analysis, penetration testing and digital forensics.
Hostel Administration	How can I reach the Chief Hostel Warden for room allotment issues?	The Chief Warden's office is at Block A Hostel Office, Room 002. Email: warden@institution.edu Phone: Ext. 5010.

Minimum credit requirem nt details	What are the total minimu m credit requirem ents to earn a B.E./B.T ech degree under Regulations 2023?	Under the regulations 2023, the credit requirements to earn a B.E./B.Tech degree is 160 to 165 credits.
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Figure 3.2.Result and Analysis

1.The figure shows the execution of the terminal assistant for the query “How is the GPA calculated under the 2023 Autonomous Regulations?.The system retrieves the relevant information from the institutional regulations and generates a source-grounded response explaining the GPA calculation method.

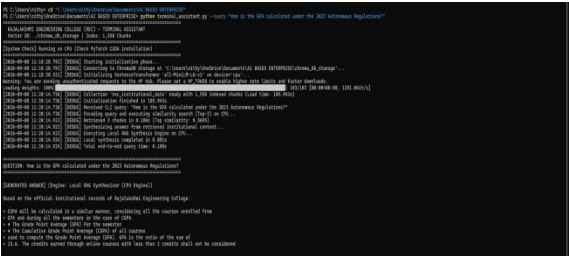


Figure 3.3

2.The figure shows the terminal execution of the proposed RAG-based assistant for identifying active Industry MoUs of the Computer Science and Engineering department. The system retrieves relevant institutional documents using similarity search and generates a source-grounded response along with document and page-level citations, demonstrating reliable and traceable information retrieval.

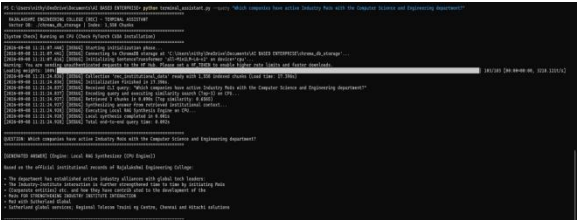


Figure 3.4

3.The figure shows the terminal execution of the proposed RAG-based assistant for retrieving UG passing marks guidelines. The system searches the institutional knowledge base, retrieves relevant academic regulations, and generates a source-grounded response describing the required passing marks and assessment criteria.

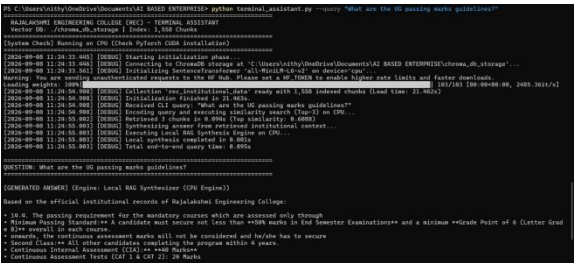


Figure 3.5

4. The figure shows the terminal execution of the proposed RAG-based assistant for retrieving the minimum attendance requirement for end-semester examinations. The system searches the institutional knowledge base, retrieves relevant regulatory information, and generates a source-grounded response based on the official college records.

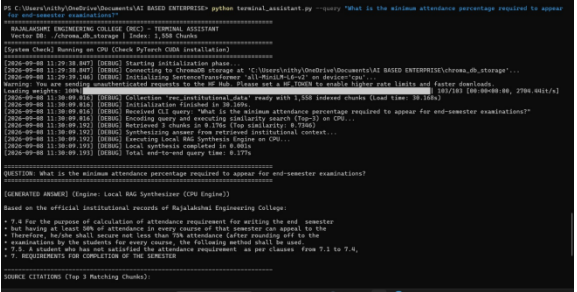


Figure 3.6

5. The figure shows the terminal execution of the proposed RAG-based assistant for retrieving

[illegible]

VII.CONCLUSION

provides a centralized, intelligent gateway for navigating campus administrative and faculty records. By uniting Natural Language Processing, vector embeddings, semantic search, and Retrieval-Augmented Generation, the platform allows students to get instant answers using everyday language. This streamlines campus navigation, eliminating the hassle of digging through fragmented sources or constantly interrupting staff for routine details.

A student asks, “Who is the faculty coordinator for the CSE department?”

The assistant provides the faculty coordinator's name and relevant contact details from the college records.

A student asks, “Where is the Exam Cell located?”

The assistant gives the correct building, room, or office location of the Exam Cell.

A student asks, “Where can I find information about college regulations?”

The assistant provides the relevant regulation details from the uploaded college documents.

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